TRANSFORMATION OF PARAMETERS

Let $p_{\alpha,\beta}(\alpha,\beta)$ denote the prior for (α,β). We wish to find the prior $p_{\gamma,\delta}(\gamma,\delta)$ where the transformed parameters satisfy

$$\gamma = \log(\alpha/\beta) \qquad \delta = \log(\alpha+\beta)$$

In general, when $(\gamma,\delta) = g(\alpha,\beta)$, the recipe for finding the distribution of (γ,δ) is:

1. Find the inverse transformation $g^{-1}(\gamma,\delta) = (\alpha,\beta)$.
2. Compute the Jacobian of the inverse transformation:
$$J = \begin{pmatrix} \frac{\partial\gamma}{\partial\alpha} & \frac{\partial\gamma}{\partial\beta} \\ \frac{\partial\delta}{\partial\gamma} & \frac{\partial\delta}{\partial\gamma} \end{pmatrix}$$
3. Compute the determinant of the Jacobian
$$\det(J) = \begin{Vmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{Vmatrix} = J_{11}J_{22} - J_{12}J_{21}.$$
4. Finally, the pdf of (γ,δ) is
$$p_{\gamma,\delta}(\gamma,\delta) = p_{\alpha,\beta}\big(g^{-1}(\gamma,\delta)\big)\,|J|$$

In the present case, the inverse transformation is

$$\alpha = \frac{e^{\gamma+\delta}}{1+e^{\gamma}} \qquad \beta = \frac{e^{\delta}}{1+e^{\gamma}}$$

so the determinant of the Jacobian of the inverse transformation is

$$\det(J) = \begin{Vmatrix} \frac{\partial\alpha}{\partial\gamma} & \frac{\partial\beta}{\partial\gamma} \\ \frac{\partial\alpha}{\partial\delta} & \frac{\partial\beta}{\partial\delta} \end{Vmatrix} = \begin{Vmatrix} \frac{e^{\gamma+\delta}}{(1+e^{\gamma})^2} & \frac{e^{\gamma+\delta}}{1+e^{\gamma}} \\ -\frac{e^{\gamma+\delta}}{(1+e^{\gamma})^2} & \frac{e^{\delta}}{1+e^{\gamma}} \end{Vmatrix}$$

$$= \frac{e^{\gamma+\delta}}{(1+e^{\gamma})^2}\frac{e^{\delta}}{1+e^{\gamma}} + \frac{e^{\gamma+\delta}}{1+e^{\gamma}}\frac{e^{\gamma+\delta}}{(1+e^{\gamma})^2} = \frac{e^{\gamma+\delta}}{1+e^{\gamma}}\frac{e^{\delta}}{1+e^{\gamma}}.$$

The determinant is always positive, so the prior for (γ,δ) is

$$p_{\gamma,\delta}(\gamma,\delta) = p_{\alpha,\beta}\left(\frac{e^{\gamma+\delta}}{1+e^{\gamma}}, \frac{e^{\delta}}{1+e^{\gamma}}\right)\frac{e^{\gamma+\delta}}{1+e^{\gamma}}\frac{e^{\delta}}{1+e^{\gamma}}$$

Notice that $J = \alpha\beta$ so if $p_{\alpha,\beta}(\alpha,\beta) = (\alpha+\beta)^{-5/2}$, we can write this as

$$p_{\gamma,\delta}(\gamma,\delta) = p_{\alpha,\beta}(\alpha,\beta)\alpha\beta = (\alpha+\beta)^{-5/2}\alpha\beta.$$

You can save some effort by using $1/\det(J) = \det(J^{-1}) = \det(\partial(\gamma,\delta)/\partial(\alpha,\beta))$.